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Database Foundations

2-2

Conceptual and Physical Data Models

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Objectives

- This lesson covers the following objectives:
  - Describe a conceptual data model
  - Describe a logical data model
  - Describe a physical data model
  - Analyze the similarities and differences between conceptual and physical data models

What Is a Conceptual Model?

- Captures the functional and informational needs of a business
- Is based on current needs, but may reflect future needs
- Addresses the needs of a business (what is conceptually ideal), but does not address its implementation (what is physically possible)

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What Is a Conceptual Model?

- Identifies :
  - important entities (objects that become tables in database)
  - relationships among entities
- Does not specify :
  - attributes (objects that become columns or fields in database)
  - unique identifiers (attribute that becomes primary key in database)

What Is a Logical Model?

- Includes all entities and relationships among them
- Is called an entity relationship model (ERM)
- Is illustrated in an ERD
- Specifies all attributes and UIDs for each entity
- Determines attribute optionality
- Determines relationship optionality and cardinality *

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A logical data model describes the data in as much detail as possible, without regard to how it will be physically implemented in the database. It is normally derived from a conceptual data model.

* We will discuss optionality and cardinality later in the course

What Is a Physical Model?

- Is an extension to a logical data model
  - Defines table definitions, data types, and precision
  - Identifies views, indexes, and other database objects *
- Describes how the objects should be implemented in specific database
- Shows all table structures, including columns, primary keys, and foreign keys

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Physical modeling deals with the conversion of the logical data model to a relational database model. Each relational model can have one or more physical models, one for each RDBMS that you deploy to.

* We will discuss views, indexes and other database objects later in the course.

Steps to Create a Physical Data Model

1. Model entities as tables
2. Model relationships as foreign keys
3. Model attributes as columns
4. Modify the physical data model based on physical constraints and requirements

Conceptual and Physical Models

- The art of planning, developing, and communicating produces a desired outcome
- Data modeling is the process of capturing the important concepts and rules that shape a business and depicting them visually in a diagram
- This diagram becomes the blueprint for designing the physical thing
- The client's dream (conceptual model) becomes a physical reality (physical model)

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The conceptual model is concerned with the real-world view and understanding of data. The physical model specifies how it will be executed in a particular database management instance.

Conceptual/Logical Model: Case Scenario

Matt, I would like you to create a simplified library database to store the book details in our department

Faculty

Sure. I will start by identifying the entities and attributes and their relationships

Matt

Case Scenario: Creating a Conceptual Model

- A Conceptual data model documents important entities and how they relate to each other

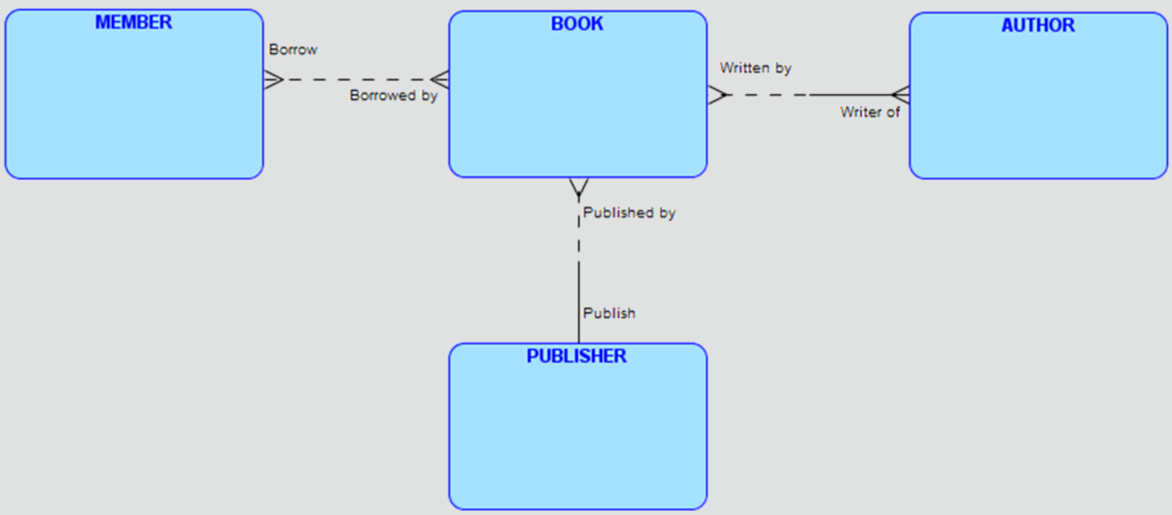

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Diagram was created using Oracle SQL Data Modeling tool which will be covered in Section 4 of course material.

Case Scenario: Creating a Logical Model

- A logical data model documents the information requirements of the business

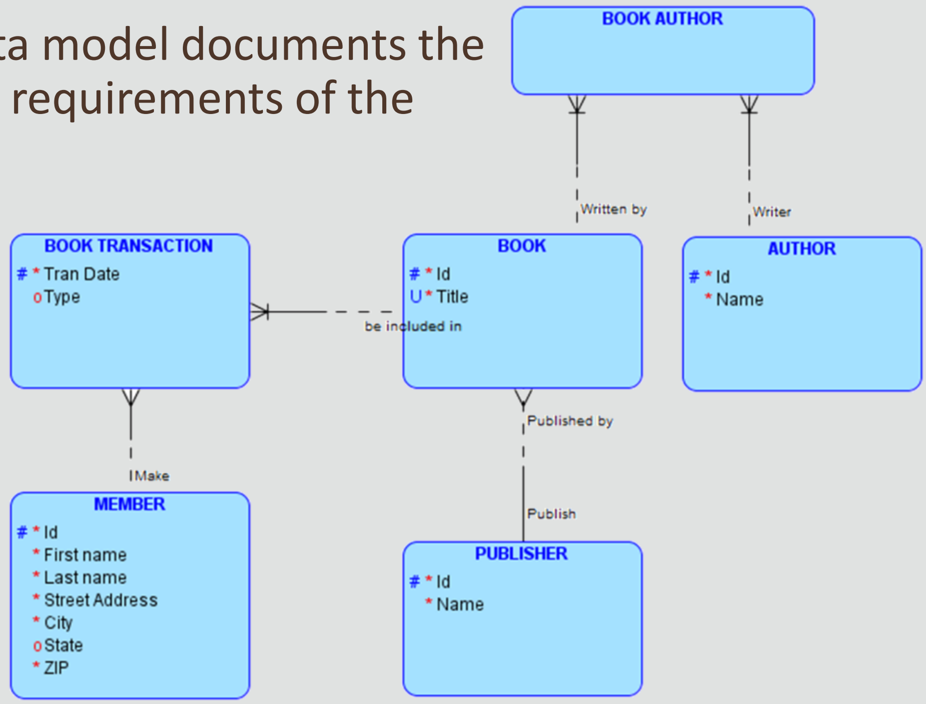

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Diagram was created using Oracle SQL Data Modeling tool which will be covered in Section 4 of course material.

Physical Model: Case Scenario

Faculty

Matt, I want to know the specifications for all tables and columns required in the simplified library database

Sure. I will convert the entities and their attributes into tables and columns

Matt

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Case Scenario: Creating a Physical Model

Modeling Performed :

Entities ⟶ Tables
Attributes ⟶ Columns
Relationships ⟶ Foreign keys

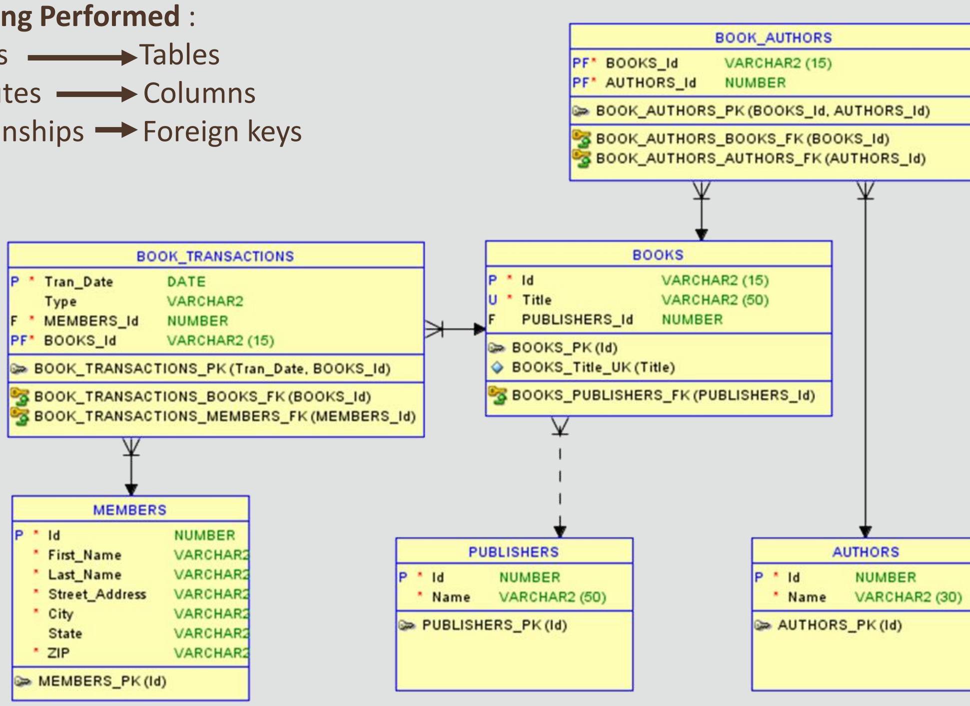

Diagram was created using Oracle SQL Data Modeling tool which will be covered in Section 4 of course material.

Summary

- In this lesson, you should have learned how to:
  - Describe a conceptual model
  - Describe a logical model
  - Describe a physical model
  - Analyze the similarities and differences between conceptual and physical data models

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